Week 3

3.1 Continuity

Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$, $\vec{a} \in A$. We give two different definitions for the function f being continuous at the point $\vec{a}$. The first is an $\epsilon - \delta$ definition, whereas the second uses the notion of a limit.

Definition 3.14. *f is continuous at $\vec{a}$ if*

- $\forall \epsilon > 0$, $\exists \delta > 0$ *such that, if $\vec{x} \in A$ and $\|\vec{x} - \vec{a}\| < \delta$, then $|f(\vec{x}) - f(\vec{a})| < \epsilon$.*
- $\lim_{\vec{x}\to\vec{a}} f(\vec{x})$ *exists and equals $f(\vec{a})$.*

f is continuous on A if f is continuous at $\vec{a}$, for every $\vec{a} \in A$.

Example 60. Fix $k \in \{1, \dots, n\}$, and let $F_k : \mathbb{R}^n \to \mathbb{R}$ denote the coordinate function

$$F_k(\vec{x}) = F_k((x_1, \dots, x_n)) = x_k.$$

Fix $\vec{a} \in \mathbb{R}^n$ and $\epsilon > 0$. We then choose $\delta = \epsilon > 0$. If $\|\vec{x} - \vec{a}\| < \delta$, then Fix $\vec{a} \in \mathbb{R}^n$ and $\epsilon > 0$. We then choose $\delta = \epsilon > 0$. If $\|\vec{x} - \vec{a}\| < \delta$, then

$$|F_k(\vec{x}) = F_k(\vec{a})| = |x_k - a_k| \le \sqrt{\sum_{i=1}^{n} (x_i - a_i)^2} = \|\vec{x} - \vec{a}\| < \delta = \epsilon.$$

Therefore, F_k is continuous at $\vec{a} \in \mathbb{R}^n$, and hence F_k is continuous on $\mathbb{R}^n$ for every $k = 1, \dots, n$.

The following theorem follows directly from the properties of limits.

Theorem 61. *If $f, g : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ are continuous functions at $\vec{a} \in \Omega$, then*

1. *$f + g$, λf, fg are continuous at $\vec{a}$ ($\lambda \in \mathbb{R}$).*
2. *$\frac{f}{g}$ is continuous at $\vec{a}$, provided $g(\vec{a}) \neq 0$.*

Combining this theorem with the previous example of the coordinate function being continuous, we see that all polynomials and rational functions are continuous

Example 62. The function $f(x, y, z) = x^3 + 3yz + z^2 - x + 7y$ is continuous on $\mathbb{R}^3$, and the function $g(x, y, z) = \frac{x^3+y^3+yz}{x^2+y^2}$ is continuous on $\mathbb{R}^3 \setminus \{x = y = 0\}$.

Given a rational function $Q(x) = \frac{P_1(x)}{P_2(x)}$ for some polynomials P_1, P_2, we see that Q is continuous away from the zero level set of P_2. That is, $Q : \mathbb{R}^n \setminus \{P_2 = 0\} \to \mathbb{R}$ is continuous.

Suppose $P_2(a) = 0$. Then Q can be extended to a continuous function at a if and only if the limit $\lim_{x\to a} Q(x)$ exists.

Example 63. If $Q(x,y) = \frac{xy+y^3}{x^2+y^2}$, then Q is continuous on $\mathbb{R}^2 \setminus \{0\}$. Fix $m \geq 0$ and consider

$$\lim_{(x,mx)\to(0,0)} Q(x,mx) = \lim_{x\to 0} \frac{mx^2 + m^3x^3}{x^2 + m^2x^2} = \lim_{x\to 0} \frac{m + m^3x}{1+m^2} = \frac{m}{1+m^2}.$$

Since this varies in m, the limit $\lim_{(x,y)\to(0,0)} Q(x,y)$ DNE, and hence Q cannot be extended to a continuous function on all of $\mathbb{R}^2$.

Example 64. If $Q(x,y) = \frac{x^4-y^4-5x^2y^2}{x^2+y^2}$, then Q is continuous on $\mathbb{R}^2 \setminus \{0\}$. We note that

$$\lim_{(x,y)\to(0,0)} Q(x,y) = \lim_{r\downarrow 0} \frac{r^4(\cos^4\theta - \sin^4\theta - 5\sin^2\theta\cos^2\theta)}{r^2} = \lim_{r\downarrow 0} r^2 \underbrace{(\cos^4\theta - \sin^4\theta - 5\sin^2\theta\cos^2\theta)}_{\text{bounded in absolute value by 7}} = 0,$$

where in the final equality we have used the Squeeze theorem. Therefore Q can be extended to a continuous function on all of $\mathbb{R}^2$, by setting

$$Q(x,y) := \begin{cases} \frac{x^4-y^4-5x^2y^2}{x^2+y^2} & : (x,y) \neq (0,0) \\ 0 & : x = y = 0. \end{cases}.$$

Theorem 65. *If $f : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ is continuous at $\vec{a} \in \Omega$ and $g : \mathbb{R} \to \mathbb{R}$ is continuous at $f(\vec{a}) \in \mathbb{R}$, then $g \circ f : \Omega \to \mathbb{R}$ is continuous at $\vec{a}$, and hence*

$$\lim_{\vec{x}\to\vec{a}} g\circ f(\vec{x}) = g\left(\lim_{\vec{x}\to\vec{a}} f(\vec{x})\right) = g\circ f(\vec{a}).$$

Proof. Fix $\epsilon > 0$. Since g is continuous, there exists $\eta > 0$ such that for $y \in \mathbb{R}$, if $|y - f(\vec{a})| < \eta$, then $|g(y) - g\circ f(\vec{a})| < \epsilon$. Then, since f is continuous, there exists $\delta > 0$ such that, for $x \in \Omega$, if $\|\vec{x} - \vec{a}\| < \delta$, then $|f(\vec{x}) = f(\vec{a})| < \eta$, which then implies that $|g\circ f(\vec{x}) - g\circ f(\vec{a})| < \epsilon$ as required. ◻

Example 66. Letting $f = F_k$, the k-coordinate function from before, and $g(x) = |x|$, we see that the maps

$$(x_1,\dots,x_n) \mapsto |x_k| \text{ are continuous, for every } k = 1,\dots,n.$$

Example 67. Choosing the same f as the previous example, but $g(x) = |\log x|$, which is continuous on $(0,\infty)$, we see that for any $k \in \{1,\dots,n\}$, the map

$$(x_1,\dots,x_n) \mapsto |\log x_k| \text{ is continuous on the half-space } \{x_k > 0\}.$$

Example 68.

$$\sin(x^2+yz), \quad e^{x-y}, \quad r = \sqrt{x^2+y^2},$$

are all continuous functions everywhere in their domains.

3.2 Partial Derivatives

We now consider the rate of change of a function with respect to each variable individually.

Definition 3.15. *Let $f : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ with Ω open. For $i \in \{1, \dots, n\}$, we define the i^{th}-partial derivative of f at $\vec{x} \in \Omega$ to be*

$$\frac{\partial f}{\partial x_i}(\vec{x}) = \lim_{h \to 0} \frac{f(\vec{x} + h\vec{e}_i) - f(\vec{x})}{h},$$

where $\vec{e}_i = (0, \dots, 1, \dots, 0) \in \mathbb{R}^n$ is the unit vector with all components equal to zero except for the i^{th}-component, which is 1.

Example 69. For the function $f(x, y) = x^2 + y^2$, we have

$$\frac{\partial f}{\partial x} = 2x \quad \text{(regard } y \text{ as a constant)}$$
$$\frac{\partial f}{\partial y} = 2y \quad \text{(regard } x \text{ as a constant)}$$

We note that, at the point $(1, -1)$, we have $\frac{\partial f}{\partial x}(1, -1) = 2 > 0$, $\frac{\partial f}{\partial y}(1, -1) = -2 < 0$, so

$$f \text{ is increasing as } x \text{ increases at } (1, -1),$$
$$f \text{ is decreasing as } y \text{ increases at } (1, -1).$$

Example 70. For the function $g(x, y, z) = xy^2 - \cos(xz)$, we have

$$\frac{\partial g}{\partial x} = g_x = y^2 + z\sin(xz)$$
$$\frac{\partial g}{\partial y} = g_y = 2xy$$
$$\frac{\partial g}{\partial z} = g_z = x\sin(xz)$$

Example 71. In this example we consider the function

$$f(x, y) = \begin{cases} 1 & : xy \geq 0 \\ 0 & : xy < 0 \end{cases}.$$

To calculate the partial derivative f_x, we fix $y \in \mathbb{R}$ and differentiate with respect to x:

- Fix $y = 1$. Then
$$f(x, 1) = \begin{cases} 1 & : x \geq 0 \\ 0 & : x < 0 \end{cases},$$
and hence $f_x(1, 1) = 0$, $f_x(0, 1)$ DNE.

- Fix $y = 0$. Then $f(x, 0) \equiv 1$ and hence $f_x(x, 0) \equiv 0$.

Similarly, $f_y(0, 0) = 0$. However, f is **not** continuous at the origin.

The previous example shows that just because the partial derivatives exist at a point does not imply that the function is continuous at this point.

3.2.1 Higher Order Derivatives

Given a function $f : \mathbb{R}^2 \to \mathbb{R}$, there are exactly two first order partial derivatives f_x, f_y. However, we may then take partial derivatives of these functions and find four different second order partial derivatives of f:

$$\begin{aligned}
\frac{\partial^2 f}{\partial x^2} &= f_{xx} = \frac{\partial(f_x)}{\partial x}, \\
\frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial(f_x)}{\partial y} = f_{xy}, \\
\frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial(f_y)}{\partial x} = f_{yx}, \\
\frac{\partial^2 f}{\partial y^2} &= f_{yy} = \frac{\partial(f_y)}{\partial y}.
\end{aligned}$$

Repeating, we find that f has 2^k different partial derivatives of order k. e.g. f_{yyx} is a partial derivative of order three.

Example 72. Find all first and second order partial derivatives of the function $f(x, y) = x \sin y + y^2 e^{2x}$:

$$\begin{aligned}
f_x &= \sin y + 2y^2 e^{2x} \\
f_{xx} &= 4y^2 e^{2x} \\
f_{xy} &= \cos y + 4ye^{2x} \\
f_y &= x \cos y + 2ye^{2x} \\
f_{yx} &= \cos y + 4ye^{2x} \\
f_{yy} &= -x \sin y + 2e^{2x}
\end{aligned}$$

In the previous example, $f_{xy} = f_{yx}$. It turns out however, that this is not always true.

Example 73. Let

$$f(x, y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} & : (x, y) \neq (0, 0) \\ 0 & : x = y = 0 \end{cases}.$$

For $y \neq 0$, $f(x,y) = \frac{xy(x^2-y^2)}{x^2+y^2}$ near $(0,y)$, and therefore near $(0,y)$ we have

$$f_x(x,y) = \frac{(3x^2y - y^3)(x^2+y^2) - 2x^2y(x^2-y^2)}{(x^2+y^2)^2}.$$

Substituting $x = 0$ into the above equation, we conclude that

$$f_x(0,y) = \frac{-y^5}{y^4} = -y, \quad \forall y \neq 0.$$

Alternatively, we have that

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0,$$

and hence $f_x(0,y) = -y$ for every $y \in \mathbb{R}$, from which we deduce that $f_{xy}(0,0) = -1$.

Alternatively, f has asymmetry about the line $y = x$. That is

$$f(b,a) = -f(a,b), \quad \forall (a,b) \in \mathbb{R}^2.$$

In particular,

$$f_y(b,a) = \lim_{h\to 0} \frac{f(b,a+h) - f(b,a)}{h} = \lim_{h\to 0} -\frac{f(a+h,b) - f(a,b)}{h} = -f_x(a,b),$$

and similarly, $f_{yx}(b,a) = -f_{xy}(a,b)$. Substituting in $a = b = 0$, we find that $f_{yx}(0,0) = -f_{xy}(0,0) = -1$.

This example shows $f_{xy} \neq f_{yx}$ in general.

3.2.2 Clairaut's Theorem

The following theorem gives a sufficient condition on f for these mixed partial derivatives to agree.

Theorem 74 (Clairaut's Theorem)**.** *Let $\Omega \subseteq \mathbb{R}^2$ open, and $f : \Omega \to \mathbb{R}$. If both the partial derivatives f_{xy} and f_{yx} exist and are continuous everywhere in Ω, then $f_{yx} = f_{xy}$ on Ω.*

Remark. *To prove the theorem, we actually prove a slightly stronger version that what is stated in the theorem: $\Omega \subseteq \mathbb{R}^2$ open, $f : \Omega \to \mathbb{R}$, $a \in \Omega$. If both f_{xy} and f_{yx} exist in an open ball containing a, and are continuous at a, then $f_{xy}(a) = f_{yx}(a)$.*

In order to prove Clairaut's theorem, we shall use the Mean Value Theorem, the proof of which can be found in any introductory mathematical analysis course.

Theorem 75 (Mean Value Theorem)**.** *Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). Then there exists some $c \in (a,b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b-a}.$$

Proof of Clairaut's Theorem. We may assume that $a = (0,0) \in \Omega$. Let $h, k > 0$ so that $[0,h] \times [0,k] \subseteq \Omega$. We consider the constant

$$\alpha = f(h,k) - f(0,k) - f(h,0) + f(0,0).$$

We now apply the Mean Value Theorem (MVT) in both the x and y directions separately. That is, consider the function $g : [0,h] \to \mathbb{R}$, given by $g(x) = f(x,k) - f(x,0)$. By the MVT, we have that there exists some $h_1 \in (0,h)$ such that

$$g'(h_1) = \frac{g(h) - g(0)}{h}.$$

Note that the LHS $g'(h_1) = f_x(h_1,k) - f_x(h_1,0)$, and the denominator of the RHS $g(h) - g(0) = f(h,k) - f(h,0) - f(0,k) + f(0,0) = \alpha$. Therefore, we have

$$f_x(h_1,k) - f_x(h_1,0) = \alpha h^{-1}.$$

Next, we consider the function $G : [0,k] \to \mathbb{R}$, given by $G(y) := f_x(h_1,y)$. Applying MVT again, $\exists k_1 \in (0,k)$ such that

$$f_{xy}(h_1,k_1) = G_y(k_1) = \frac{G(k) - G(0)}{k} = \frac{f_x(h_1,k) - f_x(h_1,0)}{k} = \alpha(hk)^{-1}.$$

We now repeat the process but we swap the order of derivatives. That is, there exists $(h_2,k_2) \in (0,h) \times (0,k)$ such that

$$f_{yx}(h_2,k_2) = \alpha(hk)^{-1},$$

and therefore $f_{yx}(h_2,k_2) = f_{xy}(h_1,k_1)$. If we then take $h,k \downarrow 0$, this forces $(h_1,k_1),(h_2,k_2) \to (0,0)$. We can then use the continuity of f_{xy} and f_{yx} to conclude that $f_{yx}(0,0) = f_{xy}(0,0)$. □

Definition 3.16. *Let $\Omega \subseteq \mathbb{R}^n$ be open, $f : \Omega \to \mathbb{R}$, $r \in \mathbb{N}_0$. We call f a C^r-function if all partial derivatives of f up to order r exist and are continuous on Ω.*

f is called a smooth function, or a C^∞-function if it is a C^r-function for every $r \geq 0$.

Example 76. f is a C^0-function if it is a continuous function.

Example 77. $f(x,y)$ is a C^2-function if all of the functions $f, f_x, f_y, f_{xx}, f_{xy}, f_{yx}, f_{yy}$ exist and are continuous.

Example 78. The following class of functions are smooth within their domain of existence: Polynomials, Rational functions, Exponentials, Logarithms, and Trignometric functions.

Example 79. Sums, differences, products, quotients and compositions of all of the classes of functions from the previous example are also smooth within their domain of existence. e.g. $\exp(x^2 - y)\sin(\frac{x}{y})$ is a smooth function on $\{(x,y) \in \mathbb{R}^2 : y \neq 0\}$.

The following Corollary of Clairaut's theorem holds by repeated application of the theorem to a function and its derivatives.

Corollary 2. *If $f : \Omega \to \mathbb{R}$ is a C^r-function on the open set $\Omega \subseteq \mathbb{R}^n$ for some $r \geq 0$. Then the order of taking partial derivatives does not matter for all partial derivatives up to order r.*

Example 80. If $f(x,y,z)$ is a C^3-function, then

$$f_{xz} = f_{zx}, \quad f_{xyz} = f_{zxy} = f_{yzx}, \quad f_{xxy} = f_{xyx} = f_{yxx}.$$

3.3 Differentiability

In one dimension, $f : \mathbb{R} \to \mathbb{R}$ is differentiable at $a \in \mathbb{R}$ if the following limit exists

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

In the multivariable, or higher dimensional domain situation $f : \mathbb{R}^n \to \mathbb{R}$, $\vec{a} \in \mathbb{R}^n$, why can't we use the same defintion?

$$\lim_{\vec{x} \to \vec{a}} \frac{\overbrace{f(\vec{x}) - f(\vec{a})}^{\in \mathbb{R}}}{\underbrace{\vec{x} - \vec{a}}_{\in \mathbb{R}^n}}.$$

We cannot divide a real number by a vector in $\mathbb{R}^n$! In order to formulate a suitable definition in this situation, we reconsider the single variable case again.

3.3.1 Affine Approximations

Suppose $f : \mathbb{R} \to \mathbb{R}$ is differentiable at $a \in \mathbb{R}$. Then, for x close to a, we have that

$$f(x) \approx L(x) := f(a) + f'(a)(x - a),$$

where $L(x)$ is the *best* affine function (polynomial of degree one or less) to approximate $f(x)$ about the point a.

To make this notion of best approximation more precise, let us consider the error function:

$$\begin{aligned} \varepsilon(x) &:= f(x) - L(x) \\ &= f(x) - f(a) - f'(a)(x - a). \end{aligned}$$

Since $x - a \in \mathbb{R}$, we can divide through by it to see that

$$\frac{\varepsilon(x)}{x - a} = \frac{f(x) - f(a)}{x - a} - f'(a),$$

and so by definition, taking $x \to a$ we have

$$\lim_{x \to a} \frac{\varepsilon(x)}{x - a} = f'(a) - f'(a) = 0,$$

or equivalently

$$\lim_{x \to a} \frac{\varepsilon(x)}{|x - a|} = 0. \tag{3.10}$$

That is, the error function is small compared to the distance between x and a.

Exercise: Show that this choice of affine approximation is the only one for which (3.10) holds.

In higher dimensions, the graph of f should be approximated by a higher dimensional affine object (e.g the tangent plane of $z = f(x, y)$).

Example 81. Let $f : \mathbb{R}^2 \to \mathbb{R}$ and assume $f_x(a, b)$ and $f_y(a, b)$ exist. Then we could take the affine approximation at (a, b) of f to be:

$$f(x, y) \approx L(x, y) := f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

In particular, $z = L(x, y)$ is a hyperplane touching Graph(f) at the point $(a, b, f(a, b))$.

3.3.2 The Definition of Differentiability

The above discussion leads to the following definition.

Definition 3.17. *Let $\Omega \subseteq \mathbb{R}^n$ open, $\vec{a} \in \Omega$, $f : \Omega \to \mathbb{R}$. Then, f is said to be differentiable at $\vec{a}$ if*

- *All partial derivatives $\frac{\partial f}{\partial x_i}(\vec{a})$ exist, for $i \in \{1, \dots, n\}$.*
- *Given the affine approximation of f at $\vec{a}$*

$$f(\vec{x}) = \underbrace{f(\vec{a}) + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\vec{a})(x_i - a_i)}_{L(\vec{x})} + \underbrace{\varepsilon(\vec{x})}_{\text{error}},$$

  the error term satisfies $\lim_{\vec{x}\to\vec{a}} \frac{\varepsilon(\vec{x})}{\|\vec{x}-\vec{a}\|} = 0$.

That is, f is differentiable at $\vec{a}$ if f can be well-approximated by an affine function locally about the point $\vec{a}$.

Remark.
- *$L(x)$ is a degree one or less polynomial in the variables $x_1, \dots, x_n$.*
- *$L(a) = f(a)$ and $\frac{\partial L}{\partial x_i}(a) = \frac{\partial f}{\partial x_i}(a)$.*
- *$y = L(x)$ is an n-dimensional hyperplane in $\mathbb{R}^{n+1}$ tangent to Graph(f) at $(x, f(x))$.*

Example 82. Let $f(x, y) = x^2 y$.

(i) Show that f is differentiable at $(1, 2)$.

(ii) Approximate $f(1.1., 1.9)$ using the derivative.

(iii) Find the tangent plane of $z = f(x, y)$ at the point $(1, 2, f(1, 2))$.

To show (i), we calculate the first order partial derivatives of f:

$$f_x = 2xy, \quad f_y = x^2,$$

and hence, at the point $(1, 2)$ we have $f_x(1, 2) = 4$, $f_y(1, 2) = 1$. Our affine approximation at $(1, 2)$ is then

$$\begin{aligned} L(x, y) &= f(1, 2) + f_x(1, 2)(x - 1) + f_y(1, 2)(y - 2) \\ &= 2 + 4(x - 1) + (y - 2). \end{aligned}$$

Let $\varepsilon(x, y)$ denote the corresponding error function. Since

$$\begin{aligned}
\lim_{(x,y)\to(1,2)} \frac{\varepsilon(x,y)}{\|(x,y)-(1,2)\|} &= \lim_{(x,y)\to(1,2)} \frac{x^2y - 2 - 4(x-1) - (y-2)}{\|(x,y)-(1,2)\|} \qquad (h = x-1, k = y-2)\\
&= \lim_{(h,k)\to(0,0)} \frac{(1-h)^2(2+k) - 2 - 4h - k}{\|(h,k)\|}\\
&= \lim_{(h,k)\to(0,0)} \frac{h^2k + 2hk + 2h^2}{\sqrt{h^2+k^2}}\\
&= \lim_{r\downarrow 0} \frac{r^3\cos^2\theta\sin\theta + 2r^2\cos\theta\sin\theta + 2r^2\cos^2\theta}{r}\\
&= \lim_{r\downarrow 0} r^2\cos^2\theta\sin\theta + 2r\cos\theta\sin\theta + 2r\cos^2\theta = 0,
\end{aligned}$$

where the last equality is due to the Squeeze theorem. Therefore, f is differentiable at $(1, 2)$.

For (ii), $f(1.1, 1.9) \approx L(1.1, 1.9) = 2 + 4(0.1) + 1(-0.1) = 2.3$.

Finally, the tangent plane for part (iii) is given by $z = L(x, y)$. In particular, $4x + y - z = -4$, so that the tangent plane has normal vector $(4, 1, -1)$.

Example 83. Is the function $f(x, y) = \sqrt{|xy|}$ differentiable at the origin?

Note that $f(x, 0) = 0$ for every $x \in \mathbb{R}$, and $f(0, y) = 0$ for every $y \in \mathbb{R}$. So $f_x(0, 0) = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = 0$, and similarly $f_y(0, 0) = 0$.

Therefore, if f can be approximated by an affine function at the origin, the approximation is $L(x, y) \equiv 0$, with error function $\varepsilon = f$. However,

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y)}{\|(x,y)\|} = \lim_{r\downarrow 0} \frac{\sqrt{r^2\,|\cos\theta\sin\theta|}}{r} = \lim_{r\downarrow 0} \sqrt{|\cos\theta\sin\theta|},$$

which DNE, and therefore f is **not** differentiable at the origin.

Remark. *In the previous example, along the line $y = mx$, we see that*

$$f(x, mx) = \sqrt{|m|}\,|x|\,.$$

Therefore, along the x-axis $(m = 0)$, we have

$$f(x, 0) = 0 = L(x, 0),$$

and L is a good approximation to f. However, along the line $y = x$ $(m = 1)$ we have

$$f(x, x) = |x| \neq 0 = L(x, x),$$

and L is a bad approximation to f.

In the definition of differentiability, $L(x)$ is defined using only the derivative along the coordinate axes $\frac{\partial f}{\partial x_i}$. We therefore conclude that a function f is differentiable if the information in the coordinate directions can tell you information in every direction (compare this with the previous example).

Unlike the existence of partial derivatives by itself, a function being differentiable is strong enough to imply continuity of the function at that point.

Theorem 84. *If $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\vec{a} \in \Omega$, then f is continuous at $\vec{a}$.*

Proof. Let $L(\vec{x})$ denote the affine function approximation of $f(\vec{x})$ at $\vec{a}$, and $\varepsilon = f - L$. Since f is differentiable at $\vec{a}$,

$$\lim_{\vec{x}\to\vec{a}} \varepsilon(\vec{x}) = \lim_{\vec{x}\to\vec{a}} \frac{\varepsilon(\vec{x})}{\|\vec{x}-\vec{a}\|} \cdot \lim_{\vec{x}\to\vec{a}} \|\vec{x}-\vec{a}\| = 0 \cdot 0 = 0.$$

Therefore,

$$\begin{aligned}
\lim_{\vec{x}\to\vec{a}} f(\vec{x}) &= \lim_{\vec{x}\to\vec{a}} L(\vec{x}) + \lim_{\vec{x}\to\vec{a}} \varepsilon(\vec{x}) \\
&= f(\vec{a}) + \lim_{\vec{x}\to\vec{a}} \left(\sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\vec{a})(x_i - a_i) \right) + 0 \\
&= f(\vec{a}),
\end{aligned}$$

and f is continuous at $\vec{a}$. □

Exercise: Show that the affine function $L : \mathbb{R}^n \to \mathbb{R}$ defined by

$$L(x) = \lambda + \vec{\alpha} \cdot \vec{x}, \quad \forall \vec{x} \in \mathbb{R}^n,$$

for some $\lambda \in \mathbb{R}$ and $\vec{\alpha} \in \mathbb{R}^n$ is differentiable everywhere, directly from the definition.

3.3.3 Rules for Differentiation

Theorem 85. *If $\Omega \subseteq \mathbb{R}^n$ is open, and $f, g : \Omega \to \mathbb{R}$ are differentiable at $\vec{a} \in \Omega$, and $\lambda \in \mathbb{R}$ then*

(i) $f \pm g$, λf, fg are differentiable at $\vec{a}$.

(ii) $\frac{f}{g}$ is differentiable at $\vec{a}$ provided $g(\vec{a}) \neq 0$.

(iii) If $h : A \subset \mathbb{R} \to \mathbb{R}$ is a single-variable function for some open set A, and h is differentiable at $f(\vec{a}) \in A$, then $h \circ f$ is differentiable at $\vec{a}$.

The proof of this theorem is very similar to those of one-variable (e.g see Math2050).

Example 86. Constant functions are differentiable.
Coordinate functions, e.g $x \mapsto x_k$, are differentiable.
Polynomials, e.g $4x^3y + xy^2 - xyz + z^5$, are differentiable.
Rational functions, e.g $\frac{x^3y+z}{x^2+y^2+z^2+1}$, are differentiable.

The following theorem provides a simpler way to verify differentiability for more regular functions.

Theorem 87. *Let $\Omega \subseteq \mathbb{R}^n$ be open f a C^1-function on Ω. Then f is differentiable on Ω.*

Remark. *We require all of the partial derivatives to exist and be continuous on the entire open set Ω, and not just at a single point.*

Example 88. Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is a continuous function and that f_x, f_y exist and are continuous on a small open ball $B_\epsilon(0)$ about the origin. Then f is differentiable on $B_\epsilon(0)$. In particular, f is differentiable at the origin $0 \in \mathbb{R}^2$.

Example 89. Let $f : \Omega \subseteq \mathbb{R}^3 \to \mathbb{R}$ be defined by $f(x, y, z) = xe^{x+y} - \log(x + z)$, where $\Omega = \{x + z > 0\}$ open. Since

$$\begin{aligned} f_x &= (1 + x)e^{x+y} - (x + z)^{-1} \\ f_y &= xe^{x+y} \\ f_z &= -(x + z)^{-1}, \end{aligned}$$

are all continuous functions, f is C^1 on Ω, and hence f is differentiable on Ω.

Proof. We give the proof for $n = 2$. The proof for $n \geq 3$ is exactly the same but with messier notation.

Suppose $(a, b) \in \Omega \subseteq R^2$ and $B_\delta((a, b)) \subseteq \Omega$ for some $\delta > 0$. For any point $(x, y) \in B_\delta((a, b))$, we can apply the MVT to find some k lying between b and y, and some h lying between x and a, so that

$$\begin{aligned} f(x, y) - f(a, b) &= (f(x, y) - f(x, b)) + (f(x, b) - f(a, b)) \\ &= \quad f_y(x, k)(y - b) \quad + \quad f_x(h, b)(x - a). \end{aligned}$$

Therefore, we can use the partial derivatives at these points to bound the error function in the following way

$$\begin{aligned} \frac{|\varepsilon(x, y)|}{\|(x, y) - (a, b)\|} &= \frac{\left|f(x, y) - f(a, b) - f_x(a, b)(x - a) - f_y(a, b)(y - b)\right|}{\|(x, y) - (a, b)\|} \\ &= \frac{\left|(f_y(x, k) - f_y(a, b))(y - b) + (f_x(h, b) - f_x(a, b))(x - a)\right|}{\|(x, y) - (a, b)\|} \\ &\leq \left|f_y(x, k) - f_y(a, b)\right| \cdot \frac{|y - b|}{\|(x, y) - (a, b)\|} + |f_x(h, b) - f_x(a, b)| \cdot \frac{|x - a|}{\|(x, y) - (a, b)\|} \\ &\leq \left|f_y(x, k) - f_y(a, b)\right| + |f_x(h, b) - f_x(a, b)|. \end{aligned}$$

Taking $(x, y) \to (a, b)$ will force $(x, k), (h, b) \to (a, b)$, and so by the continuity of f_x and f_y and the Squeeze theorem,

$$\lim_{(x,y)\to(a,b)} \frac{|\varepsilon(x, y)|}{\|(x, y) - (a, b)\|} = 0.$$

This is indeed the definition of f being differentiable at (a, b). ◻